

Support Vector Machines (SVM)

Support Vector Machines are supervised machine learning algorithms used for classification, regression, and outlier detection.

Advantages of SVM

- Effective in high-dimensional spaces.
- Effective when the number of dimensions is greater than the number of samples.
- Uses support vectors, making it memory efficient.
- Supports different kernel functions such as linear, polynomial, and RBF kernels.

Disadvantages of SVM

- Choosing the kernel and regularization parameters carefully is important when there are many features compared with samples.
- SVMs do not directly provide probability estimates; probability estimation requires additional computation.
- Training can become computationally expensive for very large datasets.

Applications

SVMs can be used for text classification, image classification, handwriting recognition, bioinformatics, and other supervised learning tasks.